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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 02/27/2012 |
| **PI Name:** | Carranza, Susana Ph.D. | | |
| **Project Title:** | Advanced Cookware and Techniques for Food Preparation at Reduced Pressure and Gravity | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Space Human Factors Engineering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SHFH**:Space Human Factors & Habitability (archival in 2017) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | scarranza@makelengineering.com | **Fax:** | FY |
| **PI Organization Type:** | INDUSTRY | **Phone:** | (512) 512-0718 |
| **Organization Name:** | Makel Engineering, Inc. | | |
| **PI Address 1:** | 1585 Marauder Street | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Chico | **State:** | CA |
| **Zip Code:** | 95973-9064 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | SBIR Phase II |
| **Start Date:** | 06/01/2011 | **End Date:** | 05/31/2013 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Sullivan, Thomas | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX11CB41C | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Makel Engineering and Cornell University propose to develop a galley architecture taking into account the design constraints of the space habitat, such as reduced pressure and gravity, minimize size, mass, power and crew time, while producing food with high nutritional value and enough variety, acceptable taste and texture qualities for long term crew consumption. The current design of the space habitat will have a reduced atmospheric pressure of 8 psia which is equivalent to a 16,000 foot mountain top, with oxygen enrichment to prevent hypoxia effects on the crew. The combination of reduced pressure and gravity will affect the heat and mass transfer during food processing and food preparation of the food. Whether the food system is based chiefly on bulk packaged ingredients or crops grown on site, it must minimize mass, volume, power and waste, make effective use of the limited resource of crew time, produce nutritious, highly acceptable and varied food, and integrate into the closed habitat's atmospheric control system by containing and controlling airborne particulates, water vapor and odors generated during food preparation. | | |

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| | POTENTIAL NASA COMMERCIAL APPLICATIONS: The technology developed in Phase 2 will find near term applications in ground-based analog missions, where many of the subsystems will be tested. Portions of the technology (such as individual cookware items, or environmental control system) may be incorporated in flight demonstration missions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Recent research indicates that the fumes generated in the kitchen, if not properly exhausted and treated, are rather detrimental to human health. While exhaust systems are common practice in residential kitchen, they are usually ill-equipped to treat noxious fumes, but rater focus on drawing air and trapping oil mist. Many kitchens, in particular in small urban spaces, use recirculation exhaust, which simple traps oil mist, but end up recirculating odor causing volatiles and any noxious chemicals (e.g., overheated oil, small food spills on hot surfaces, etc.). A derivation of a hood system with built-in treatment may be the development of convection ovens and microwave ovens with built-in odor control systems. These items are usually outside of the range of kitchen exhaust, but also are key sources of odors and indoor pollution. |
| **Task Progress:** | New project for FY2011. Reporting not required for this SBIR Phase 2 project. |
| **Bibliography Type:** | Description: (Last Updated: ) |